

BayeSN: Past, Present and Future

Matt Grayling, Kaisey Mandel, Ben Boyd, Erin Hayes,
Stephen Thorp, Gautham Narayan, Sam Ward



The BayeSN Model

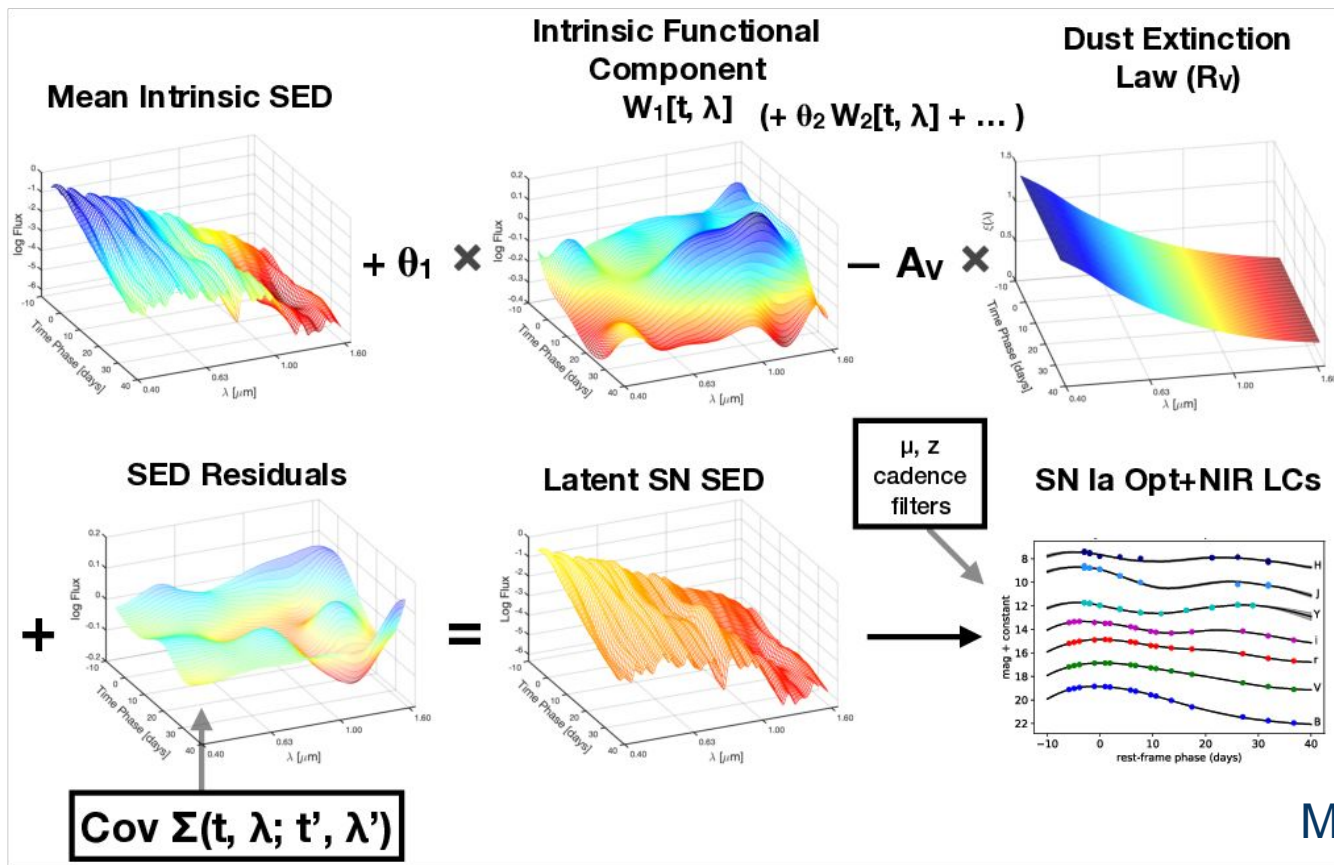
The Impact of Dust

$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

$$c^s = c_{\text{int}}^s + E(B - V)^s$$


- Colour - one parameter for two separate effects
- Correctly handling dust is key to uncover intrinsic SN properties and accurately estimate distances

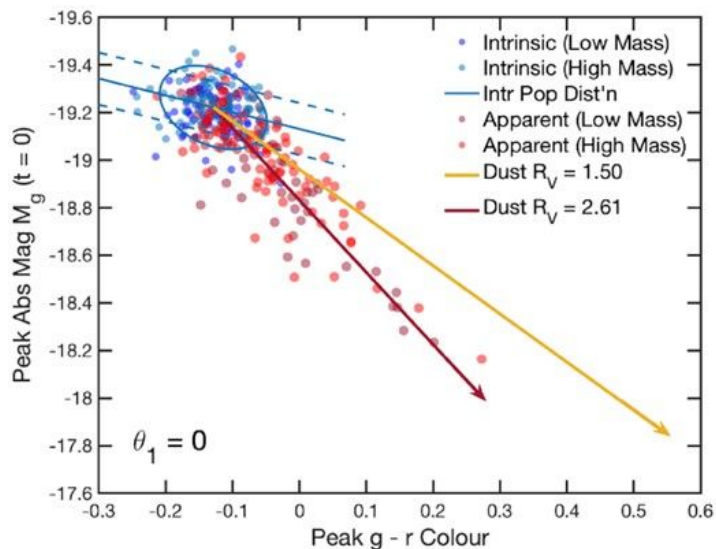
The BayeSN Model



Why use BayeSN?

Population Inference

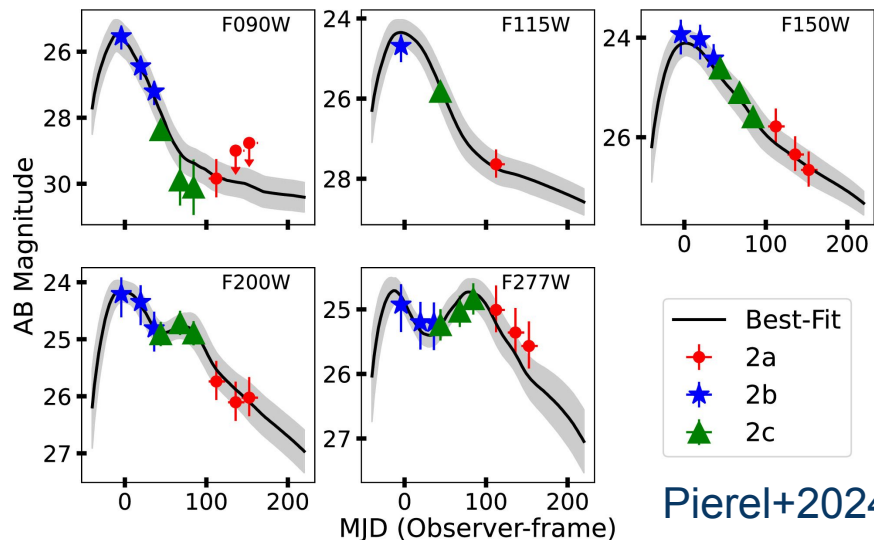
- Hierarchical Bayesian Model, constrains population properties
- Separate treatments of dust and intrinsic variation, provides astrophysical insight



Thorp
+2021

Light Curve Fitting

- Distance inferred jointly with other LC parameters, use all data rather than single colour/magnitude pairs
- Optical-NIR SED template



Pierel+2024

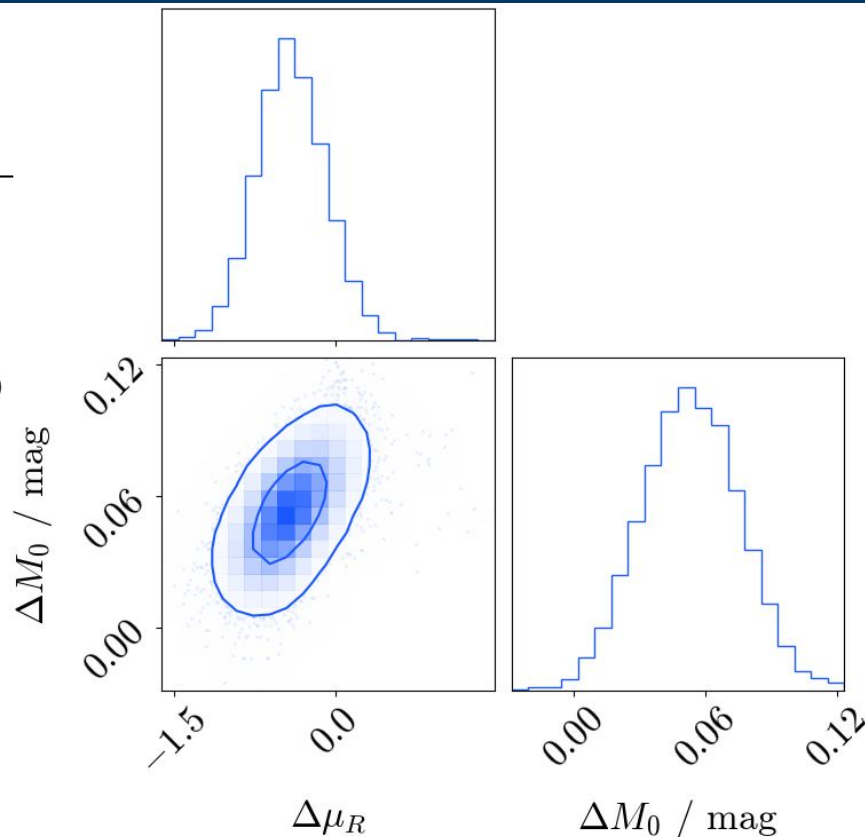
Science Results with BayeSN

Cause of the Mass Step

Sample	Intrinsic Mass Step / mag
Combined*	0.049 ± 0.019
DES5YR	0.053 ± 0.022

*Combined = DES3YR+Foundation+PS1MD

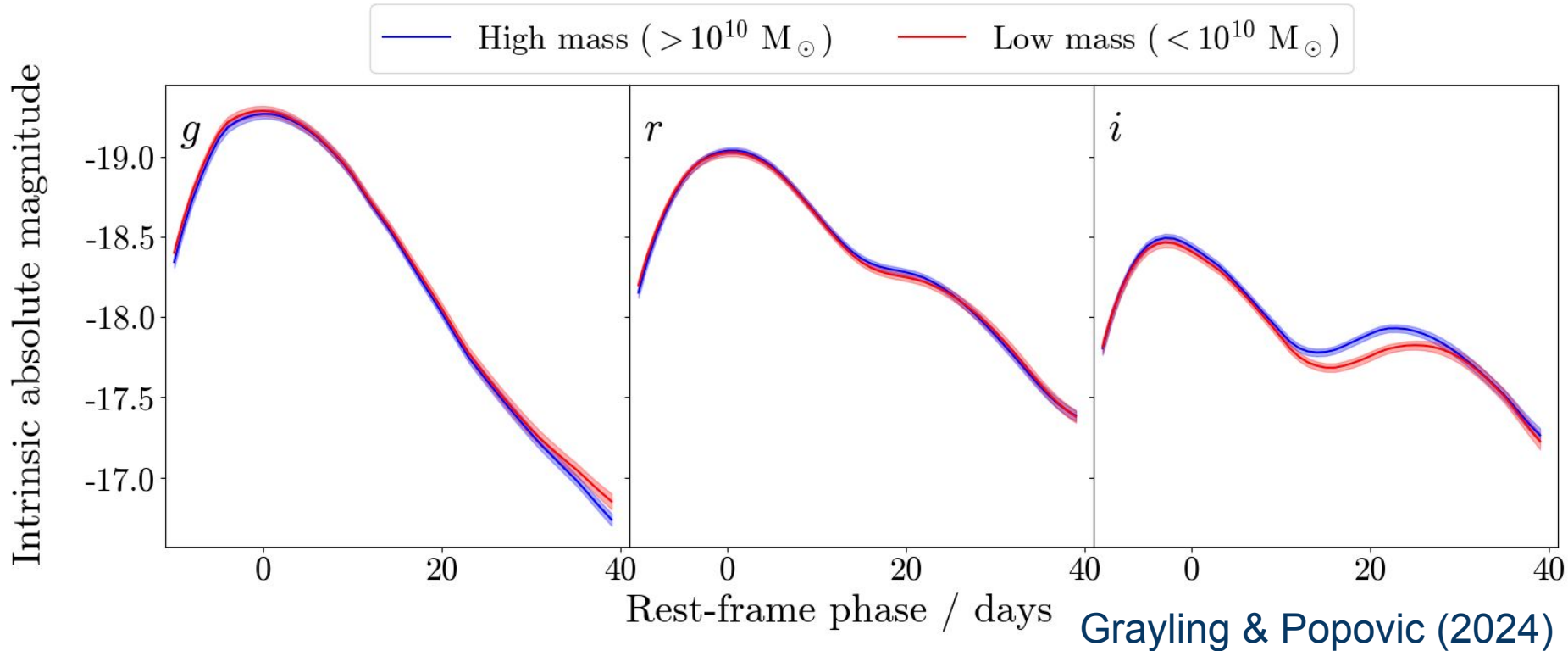
Dust may play a role, but BayeSN supports an intrinsic contribution to the mass step



Grayling & Popovic (2024)

Cause of the Mass Step

- BayeSN finds an environmental effect on *i*-band secondary maximum

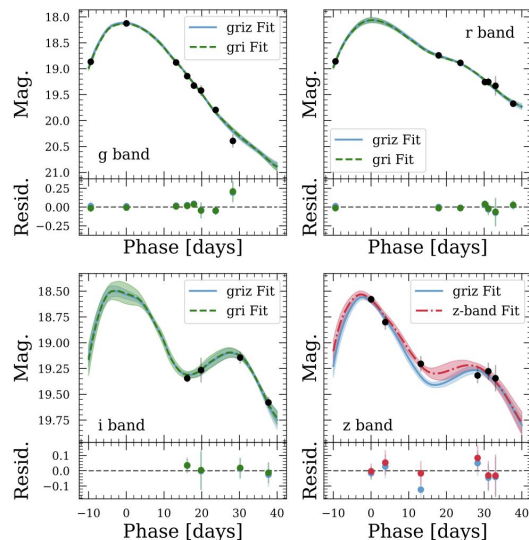


Characterising the Standardisation Properties of SNe Ia in the z band with Hierarchical Bayesian Modelling

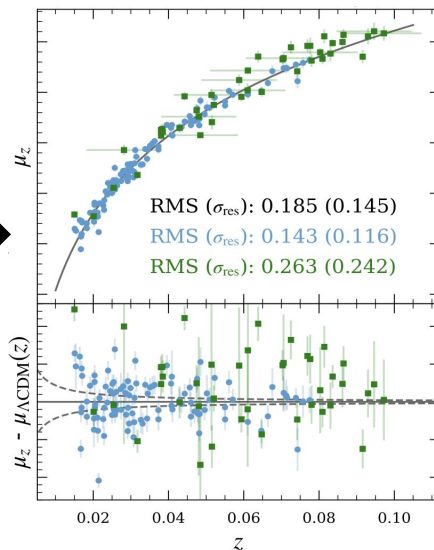
Work led by Erin Hayes (3rd Year PhD; eeh55@cam.ac.uk)



Example YSE Light Curve:
SN 2020evu



z-band Hubble Diagram



Pan-STARRS (Foundation + YSE) sample –
1st YSE cosmological analysis!

- *Pre-standardisation*, z-band residual scatter smaller than optical (*gri*)
 - z-band still correlated with optical shape parameter, θ
- Construct 1st z-band **only** Hubble diagram
- *Post-standardisation*, z-band scatter larger than in similar NIR works

Hayes et al., 2025 (submitted), on arxiv soon!

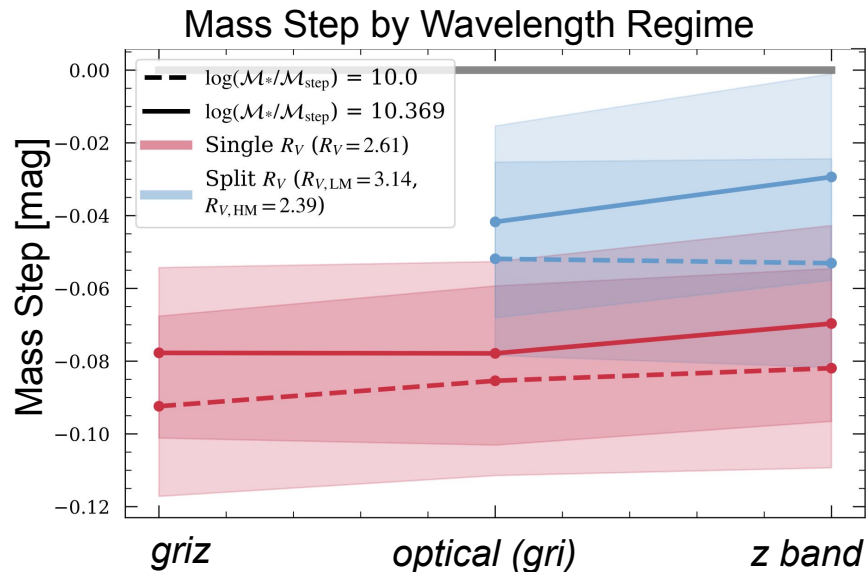
Characterising the Standardisation Properties of SNe Ia in the z band with Hierarchical Bayesian Modelling

Work led by Erin Hayes (3rd Year PhD; eeh55@cam.ac.uk)



Can dust explain the mass step (e.g. do we see a lessened mass step in the z band than optical)?

- Consistent 1-4 σ mass step in optical (*gri*) & z-band
 - Given statistical precision, no evidence that dust explains the mass step
- Result independent of assumed dust law, R_V , in high/low mass bins
 - Test with single $R_V = 2.61$ for all hosts,
 - and $R_V = 2.39/3.14$ for high/low mass hosts (values from Grayling+24)



(Above) No evidence for less significant mass step in z-band than optical across analysis choices

Hayes et al., 2025 (submitted), on arxiv soon!

The Future of BayeSN

Spectroscopic Training



Need to use spectra and photometry simultaneously

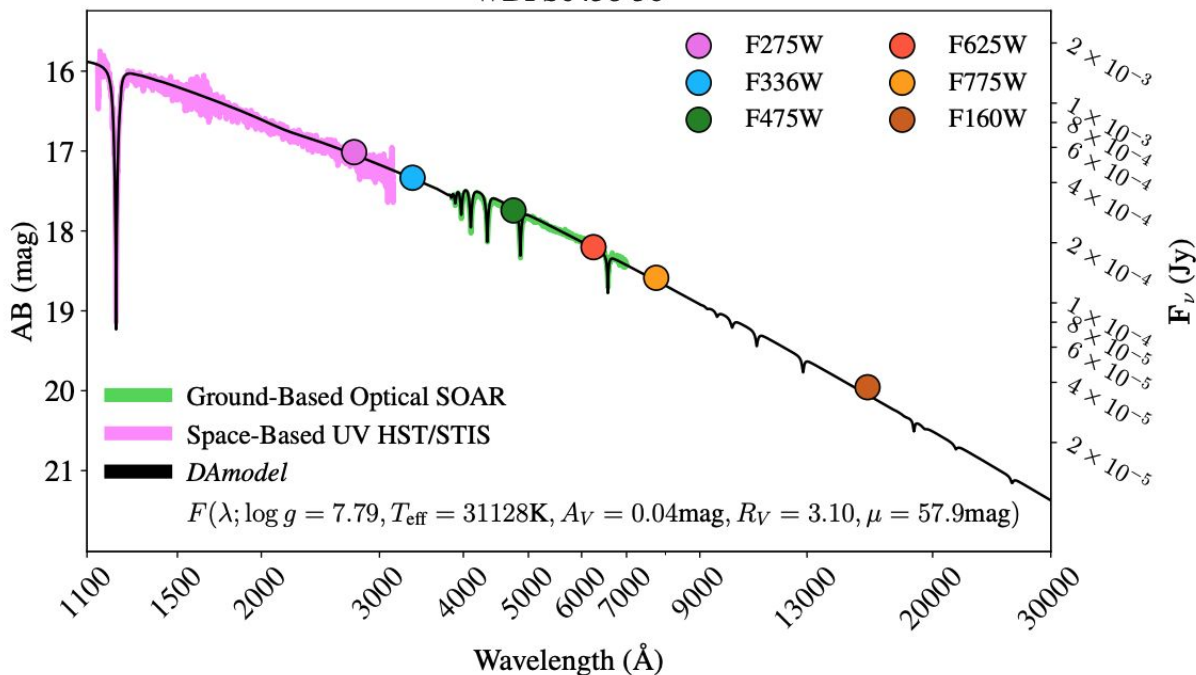
Ben has solved this problem! (See recent arxiv paper on calibration, very relevant for SN cosmology)

Future BayeSN model will be trained on spectra too!

DAmodel: Hierarchical Bayesian Modelling of DA White Dwarfs for Spectrophotometric Calibration

Benjamin M. Boyd^{1*}, Gautham Narayan², Kaisey S. Mandel^{1,3}, Matthew Grayling¹

WDFS0458-56



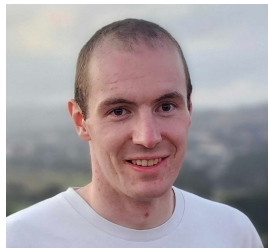
The Impact of Different Dust Laws



Aidan Reilly

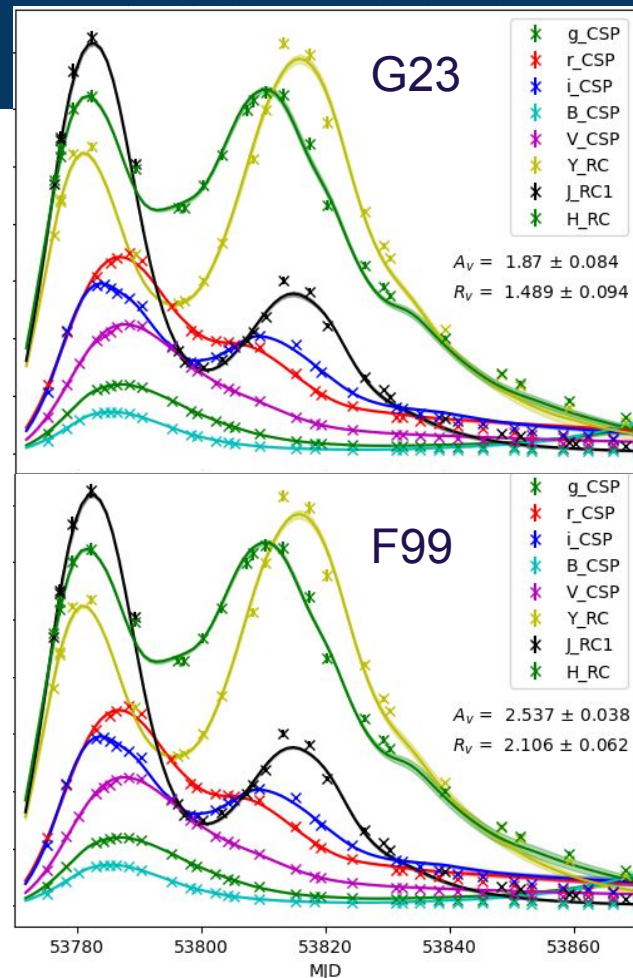


Aaron Do



Stephen Thorp

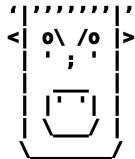
- Most of SN science using F99/CCM89 etc.
 - These dust laws are now decades old, there have been numerous updates
- Aaron Do/Stephen Thorp have done excellent work putting loads of new dust laws into BayeSN/SNANA
- Aidan Reilly leading investigation into effect of different dust laws in BayeSN, focusing on highly reddened SNe Ia



Aidan Reilly, Master's Project

Cosmology

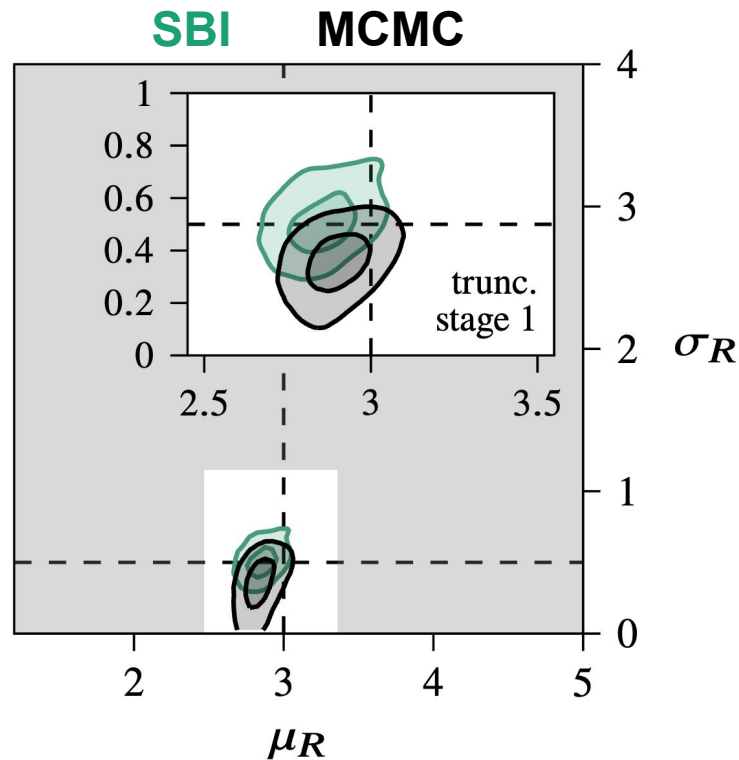
BayeSNANA



- BayeSN integrated within SNANA BBC pipeline, currently being tested
- Support from Rick Kessler, Brodie Popovic, Ayan Mitra

Simulation-based inference (SBI) with BayeSN

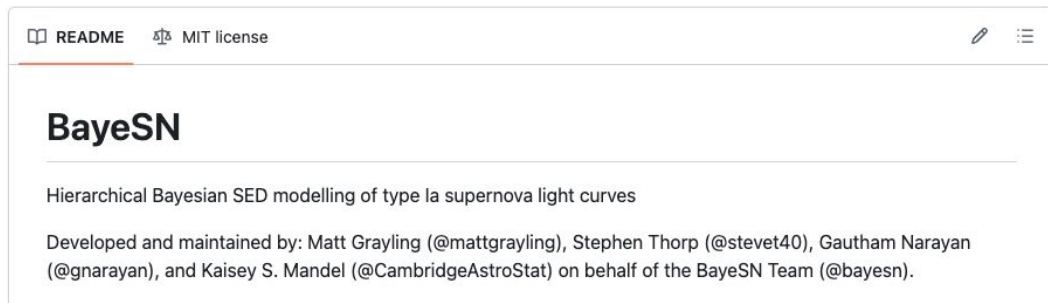
- Gaining strong traction with other cosmological probes e.g. weak lensing
- Enables single-step, hierarchical SN cosmology
- Longer term goal of BayeSN to work toward SBI cosmology
- See Karchev, Grayling et al. (2024) for an application to SN dust inference



Karchev, Grayling et al. (2024)

BayeSN Out Now!

- New GPU-accelerated, Python implementation of BayeSN out now! Use for population inference or individual light curve fitting
- github.com/bayesn/bayesn, pip installable, integrated into SNANA
- Lots of ongoing work for BayeSN e.g. training on spectra, new dust laws



MNRAS paper



New arxiv paper

Coming soon...

Cosmic Lighthouses: Astrophysical and Cosmological Challenges with SNe Ia
4-day workshop, spanning SN Ia astrophysics and cosmology

Institute of Astronomy, Cambridge

30th June - 3rd July

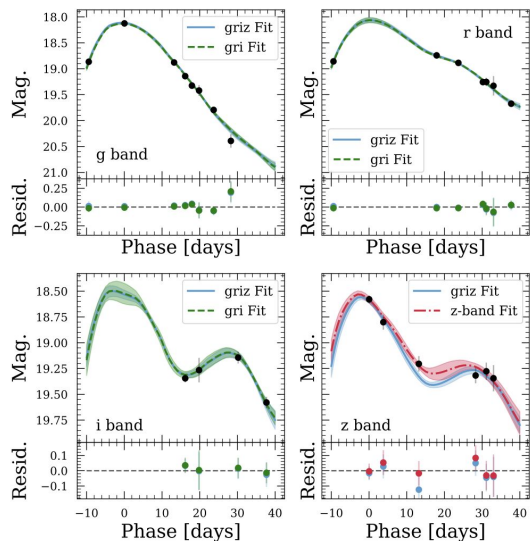
More details, formal announcement to follow...

Characterising the Standardisation Properties of SNe Ia in the z band with Hierarchical Bayesian Modelling

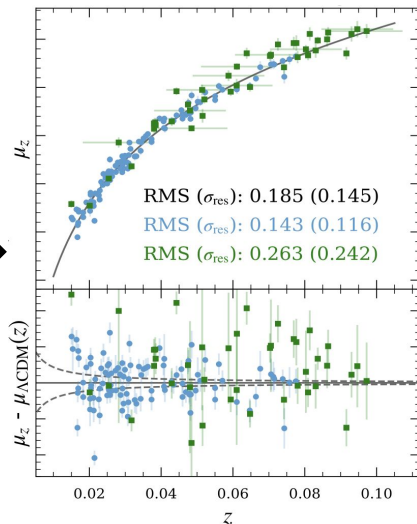
Work led by Erin Hayes (3rd Year PhD; eeh55@cam.ac.uk)



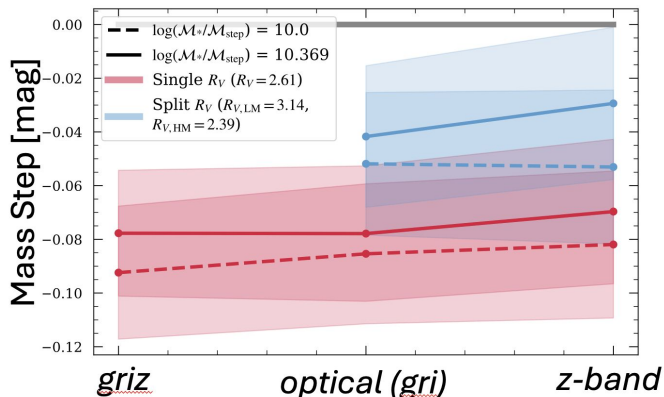
SN 2020evu



z-band Hubble Diagram



Mass Step by Wavelength Regime



- Peak z-band SN Ia mags exhibit less scatter than *gri*
- 1st z-band Hubble diagram!

Pan-STARRS (Foundation + YSE) SN Ia sample – 1st YSE cosmological analysis!

- 1-3 σ mass step in z-band that is consistent with optical (*gri*) step
- Result is independent of assumed dust law, R_V , in high/low mass bins

Full work coming soon to arxiv!